

# Semantic and Instance Segmentation

Pixel evidence becomes a class field—and, when needed, separate object masks

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ES 667 · Deep Learning · IIT Gandhinagar

**Detection drew rectangles;  
segmentation must assign  
pixels**

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# L10 returned a set of boxes; L11 changes the output support

## PUBLIC L10 · DETECTION

variable set  
(class, box, score)



## PUBLIC L11 · SEGMENTATION

class evidence at every pixel  
plus identity when objects must  
remain separate

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**The backbone still preserves spatial evidence; the head now predicts **shape**, not four box coordinates.**

# Commit: which output can answer both requirements?

QUESTION



**HELD CASE · GENERATED TEACHING SCENE** Two touching crowns form one region. Report total canopy area **and** count two trees.

**A**

one global class

**B**

one class + one  
box

**C**

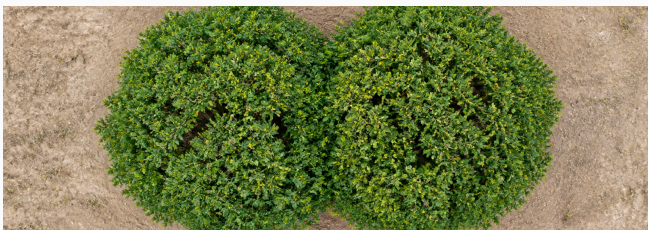
one  $H \times W$   
foreground map

**D**

a scored set of  
separate object  
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Commit silently; the same scene returns after semantic and instance outputs.

# The 80-minute route protects one causal spine

## CORE · 55 MIN

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output contract → pixel  
logits + CE → spatial  
decoder → held mask →  
IoU/Dice → Mask R-CNN  
→ evaluation + revisit

## SHOULD · 15 MIN

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void/empty conventions  
→ upsampling + RoIAlign  
→ panoptic task choice

## OPTIONAL · 10 MIN

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primary provenance →  
architecture variants →  
mask fusion / PQ  
extension

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The final one-minute checklist and L12 handoff remain CORE even when optional material is skipped.

# Three clocks prevent a probability from becoming a metric

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

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**TRAIN compares with labels; INFER makes an output; EVAL scores that output under a declared protocol.**

# The held truth is one semantic region made from two touching instances

V VISUAL

orchard tile



semantic truth  $G$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

$$|G| = 6 \text{ pixels}$$

latent identity

|   |   |   |   |
|---|---|---|---|
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| 0 | 1 | 2 | 0 |
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IDs withheld from  $G$ .

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|---|---|---|---|
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**One union supports area; two identities are required for counting.**

# Semantic, instance, and panoptic outputs answer different questions

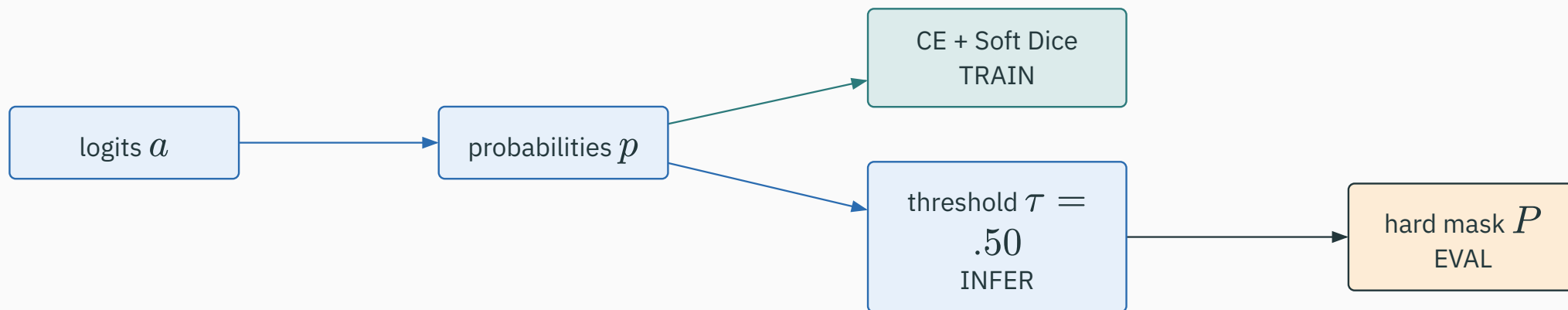
| Task     | Output contract                     | Question answered                   |
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| semantic | $H \times W$ class labels           | Which class owns this pixel?        |
| instance | variable scored set of object masks | Which pixels belong to object $j$ ? |
| panoptic | $H \times W$ class–identity pairs   | What and which object, everywhere?  |

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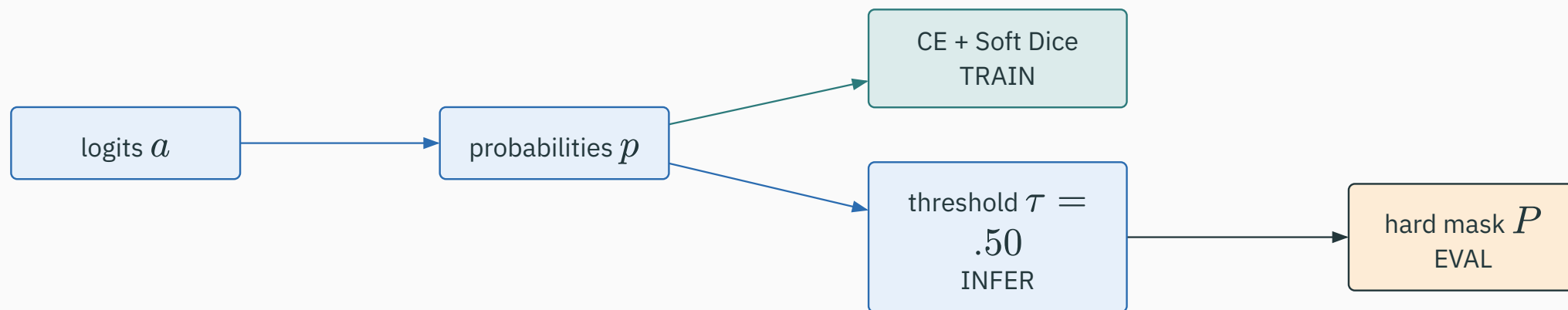
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“Stuff” such as sky usually needs a class; countable “things” such as trees also need an identity.

# One prediction will cross all three clocks



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**HELD CASE · PROMISE** The exact  $G$ ,  $p$ , threshold, and  $P$  will remain fixed through loss, inference, and evaluation.

**TRAIN: classify every spatial  
location**

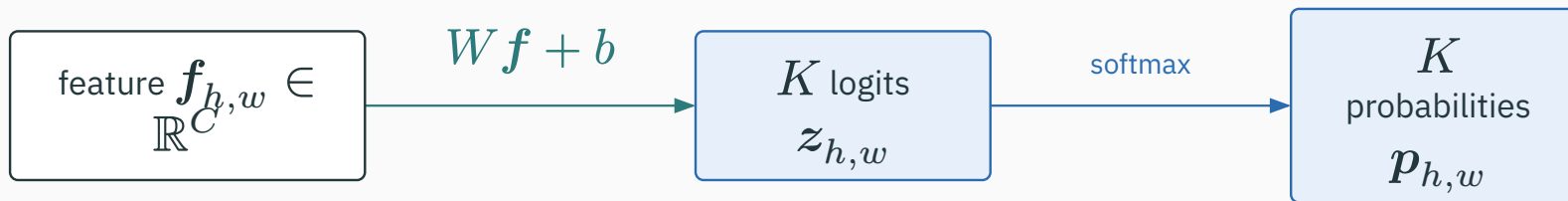
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# A semantic head emits one score vector per pixel

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**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

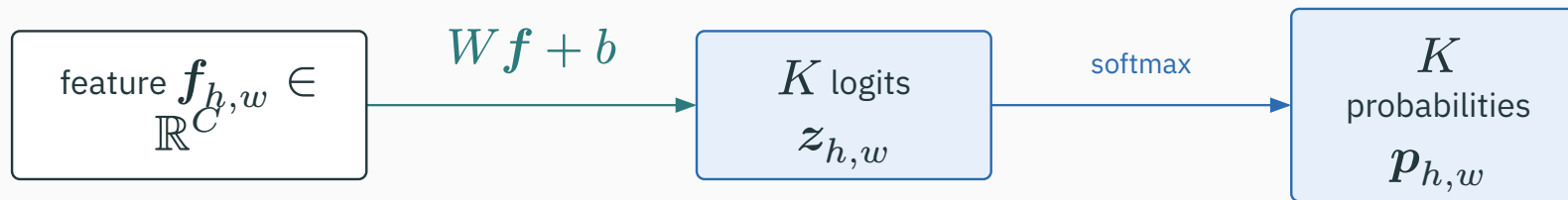


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**Segmentation shares one classifier across locations; evidence changes, weights do not. Argmax waits for INFER.**

# Softmax acts across classes at one location—not across pixels

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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**HELD CASE · ONE PIXEL** Classes are **background, road, car** and  $\mathbf{z} = \begin{pmatrix} .2 \\ 1.4 \\ -.3 \end{pmatrix}$ .

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**EXPONENTIATE**

$$\exp(\mathbf{z}) \approx \begin{pmatrix} 1.22 \\ 4.05 \\ .74 \end{pmatrix} \quad \sum_k \exp(z_k) \approx 6.01$$

**NORMALIZE**

$$\mathbf{p} \approx \begin{pmatrix} .203 \\ .674 \\ .123 \end{pmatrix} \quad p_{\text{road}} = .674$$

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$$\sum_k p_{h,w,k} = 1 \text{ independently at every } (h, w).$$

# Binary foreground can use one logit and one sigmoid

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$$p_i = \sigma(a_i) = \frac{1}{1+e^{-a_i}}, \quad a_i = \log \frac{p_i}{1-p_i}.$$

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|                       |       |      |       |        |        |
|-----------------------|-------|------|-------|--------|--------|
| $p$                   | .90   | .60  | .30   | .10    | .05    |
| $a = \text{logit}(p)$ | 2.197 | .405 | −.847 | −2.197 | −2.944 |

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The held binary mask uses one foreground probability. A  $K$ -class semantic task instead keeps  $K$  logits and softmax.

# The held foreground probabilities are now fixed

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice**INFER** probability  $\rightarrow$  label / object masks**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP**HELD CASE · PROBABILITY FIELD**  $p_i$  is the foreground probability; blue cells satisfy  $p_i \geq .50$ .truth  $G$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

foreground  $p$ 

|     |     |     |     |
|-----|-----|-----|-----|
| .10 | .90 | .90 | .10 |
| .60 | .90 | .90 | .10 |
| .10 | .90 | .30 | .60 |
| .10 | .60 | .05 | .05 |

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|     |     |     |     |
|-----|-----|-----|-----|
| .10 | .90 | .90 | .10 |
| .60 | .90 | .90 | .10 |
| .10 | .90 | .30 | .60 |
| .10 | .60 | .05 | .05 |

$$\sum_i p_i = 7.2, \quad \sum_i g_i = 6, \quad \sum_i p_i g_i = 4.8.$$

# Binary cross-entropy scores each pixel locally

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$$\ell_i = -[g_i \log p_i + (1 - g_i) \log(1 - p_i)].$$

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**FOREGROUND**  $\cdot g_i = 1$

penalty  $= -\log p_i$   
confident .90  $\rightarrow$  .105  
missed .30  $\rightarrow$  1.204

**BACKGROUND**  $\cdot g_i = 0$

penalty  $= -\log(1 - p_i)$   
safe .10  $\rightarrow$  .105  
false alarm .60  $\rightarrow$  .916

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**CE asks whether each probability is calibrated to that pixel's label.**

# Group the held pixels to reproduce dense BCE

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**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

**HELD CASE · SAME  $G, p$**  Five foreground pixels have .90; one has .30. Background has  $.60 \times 3, .10 \times 5, .05 \times 2$ .

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$$16\mathcal{L}_{\text{BCE}} = -10 \log .90 - \log .30 - 3 \log .40 - 2 \log .95$$

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$$\approx 1.0536 + 1.2040 + 2.7489 + .1026 \approx 5.1090.$$

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$$\approx 1.0536 + 1.2040 + 2.7489 + .1026 \approx 5.1090.$$

$$\mathcal{L}_{\text{BCE}} \approx \frac{5.109}{16} \approx .319.$$

# The tensor keeps spatial axes separate from the class axis

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| Axis | Size | Meaning                    |
|------|------|----------------------------|
| $h$  | $H$  | vertical location          |
| $w$  | $W$  | horizontal location        |
| $k$  | $K$  | class score or probability |

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$$Z[h, w, :] \in \mathbb{R}^K \rightarrow P[h, w, :] \in [0, 1]^K.$$

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$$Z[h, w, :] \in \mathbb{R}^K \rightarrow P[h, w, :] \in [0, 1]^K.$$

**Normalize over  $k$ ; reduce the loss over valid  $(h, w)$  locations.**

# TRAIN checkpoint: the labels supervise probabilities, not hard masks

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**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

Which object enters the derivative during training?

**A**

the thresholded mask  $P$

**B**

the differentiable  
probability  $p$

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only pixel accuracy

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**B. CE and Soft Dice operate on  $p$ ; thresholding belongs to INFER.**

**TRAIN: preserve detail while  
acquiring context**

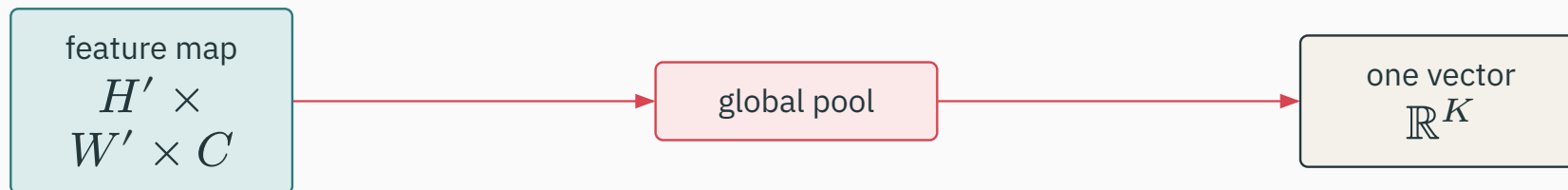
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# A classification head destroys the layout segmentation needs

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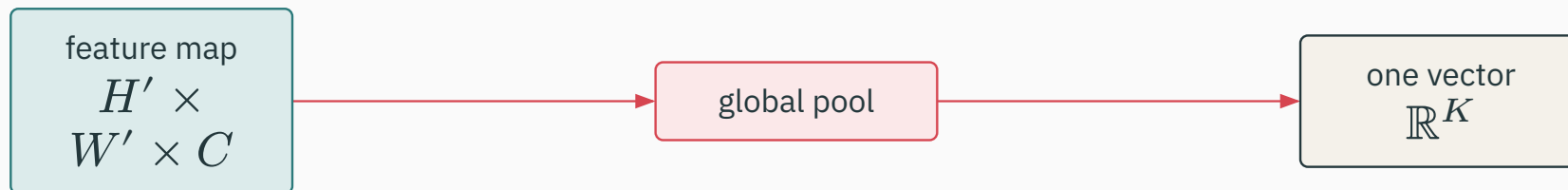


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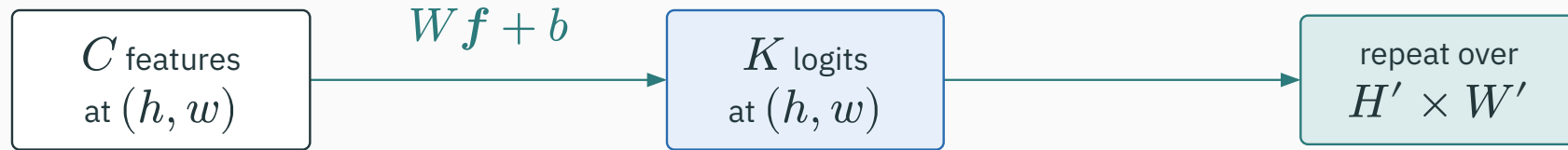
**After locations collapse, different pixels cannot receive different labels.**

# A $1 \times 1$ convolution is the shared pixel classifier

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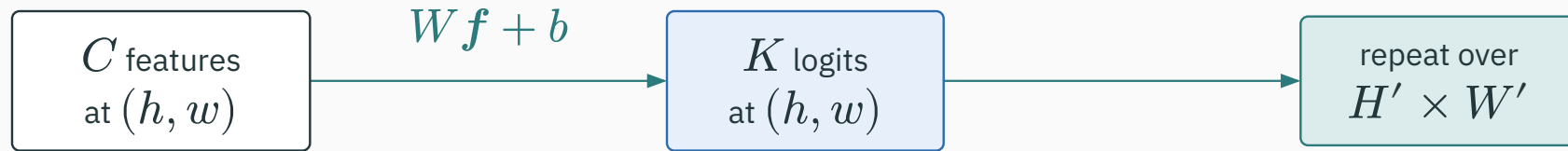


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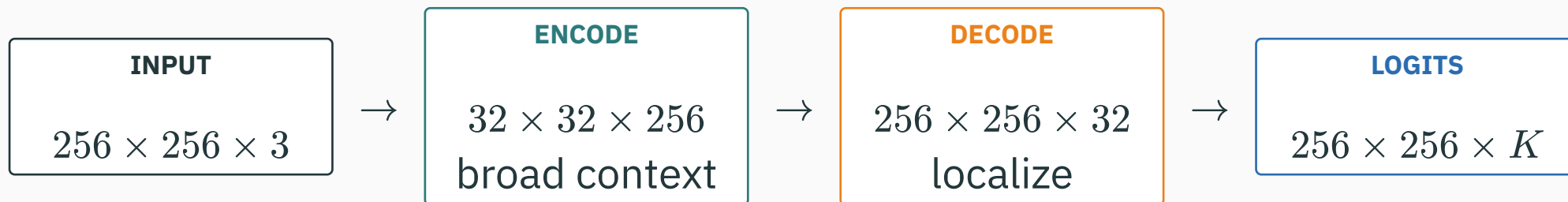
A  $1 \times 1$  kernel mixes channels at one location. Earlier spatial convolutions already gathered neighborhood evidence.

# Encoder–decoder networks exchange coordinates for context

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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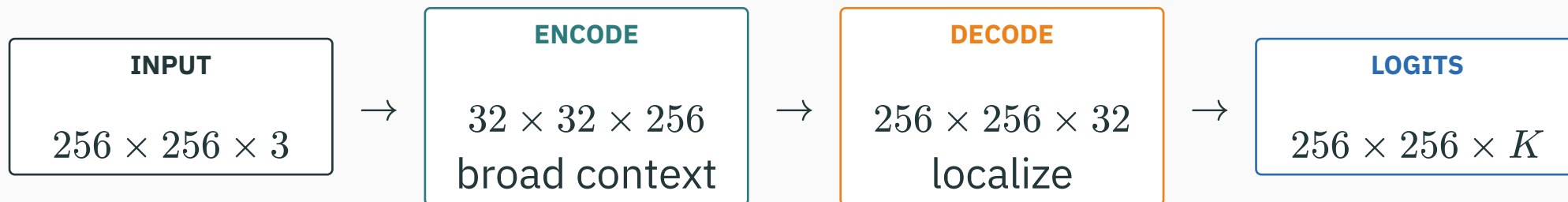


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**Downsample to understand what surrounds a pixel; upsample to decide which pixel.**

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**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

## CONTEXT GAIN

---

A deep feature can see road, kerb, and cars around one ambiguous grey patch.

## COORDINATE LOSS

---

Stride merges nearby positions; “cat” may remain correct while the ear boundary shifts.

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**Dense prediction needs deep semantics and high-resolution evidence.**

# Does a larger grid recreate discarded detail?

QUESTION

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A  $2 \times 2$  feature map is repeated into a  $4 \times 4$  grid. Did upsampling recover which source pixels created each value?

|   |   |
|---|---|
| 1 | 2 |
| 3 | 4 |



|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 2 | 2 |
| 1 | 1 | 2 | 2 |
| 3 | 3 | 4 | 4 |
| 3 | 3 | 4 | 4 |

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QUESTION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

A  $2 \times 2$  feature map is repeated into a  $4 \times 4$  grid. Did upsampling recover which source pixels created each value?

|   |   |
|---|---|
| 1 | 2 |
| 3 | 4 |



|   |   |   |   |
|---|---|---|---|
| 1 | 1 | 2 | 2 |
| 1 | 1 | 2 | 2 |
| 3 | 3 | 4 | 4 |
| 3 | 3 | 4 | 4 |

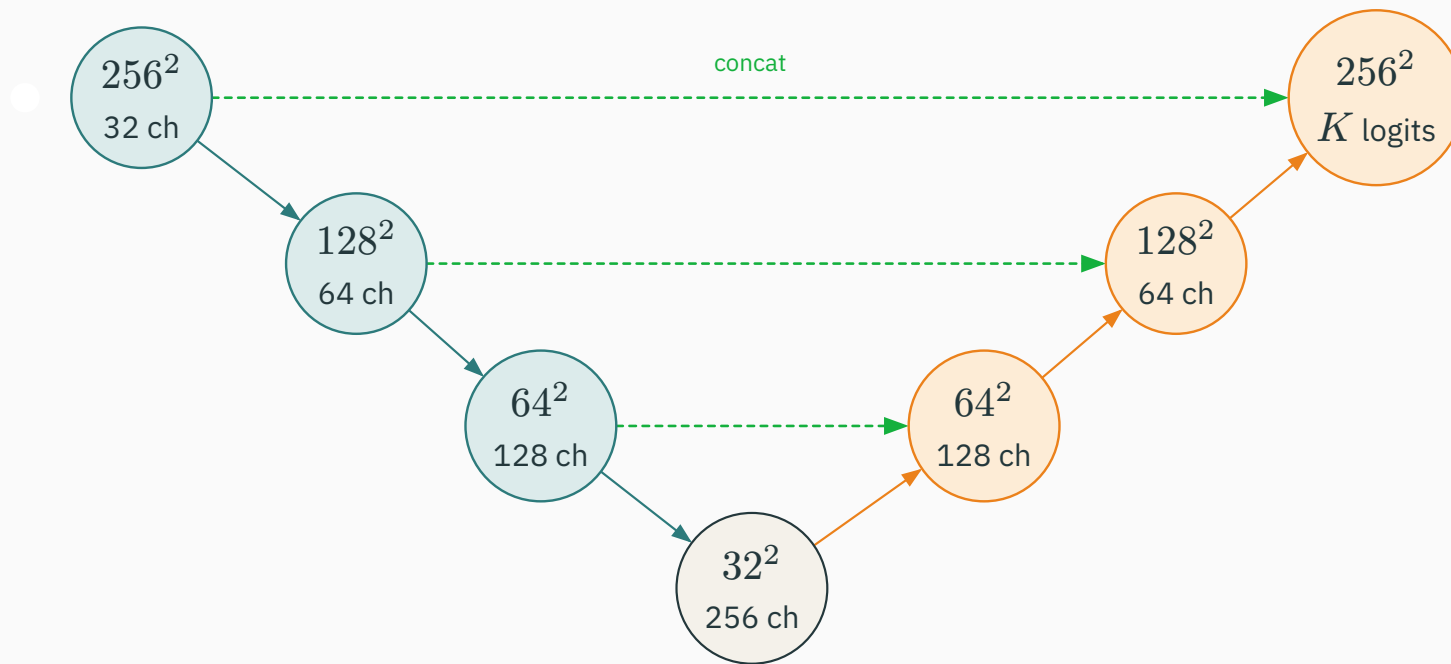
Commit before the next slide: resolution and information are not synonyms.

# U-Net routes fine detail around the bottleneck

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

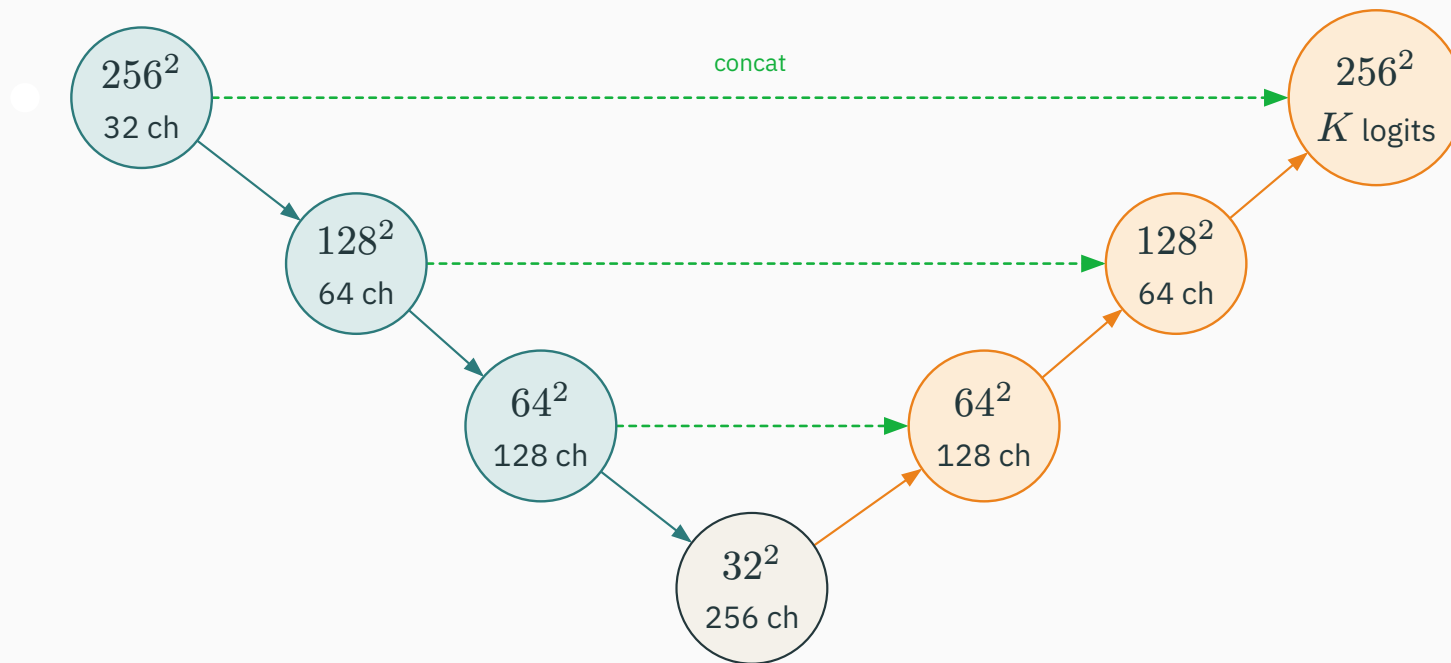


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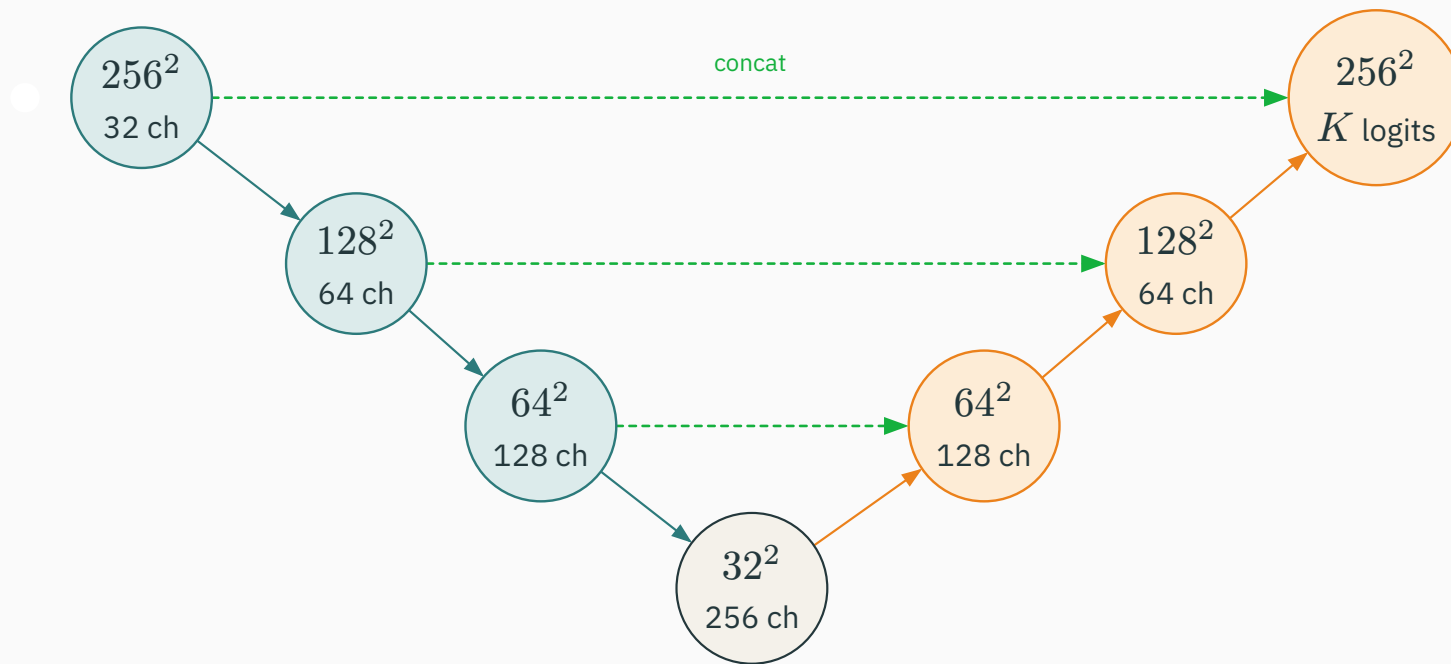
Teal learns context; orange restores resolution; green skips carry aligned fine features.

# U-Net routes fine detail around the bottleneck

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP



**Upsampling adds locations; skips restore evidence that the bottleneck discarded.**

# This ledger is a same-padding teaching U-Net—not the 2015 pixel sizes

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

## TOY LEDGER USED HERE

same-padding convolutions preserve  
each stage size; matching encoder  
and decoder tensors concatenate  
directly

## ORIGINAL U-NET

valid convolutions shrink tensors; the  
encoder feature is copied **and**  
**cropped** before concatenation

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## ORIGINAL U-NET

valid convolutions shrink tensors; the  
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$$U, E \in \mathbb{R}^{64 \times 64 \times 128} \rightarrow \text{concat}(U, E) \in \mathbb{R}^{64 \times 64 \times 256}.$$

# This ledger is a same-padding teaching U-Net—not the 2015 pixel sizes

D DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

## TOY LEDGER USED HERE

same-padding convolutions preserve each stage size; matching encoder and decoder tensors concatenate directly

## ORIGINAL U-NET

valid convolutions shrink tensors; the encoder feature is copied **and cropped** before concatenation

$$U, E \in \mathbb{R}^{64 \times 64 \times 128} \rightarrow \text{concat}(U, E) \in \mathbb{R}^{64 \times 64 \times 256}.$$

**The skip operation is concatenation in original U-Net; “skip connection” is not universally synonymous with addition.**

# Boundary failure has a smallest useful diagnostic

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice**INFER** probability  $\rightarrow$  label / object masks**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

| Symptom                        | Suspect                          | Smallest test                                 |
|--------------------------------|----------------------------------|-----------------------------------------------|
| correct class;<br>blurred edge | coarse stride or<br>missing skip | overlay boundary;<br>ablate one skip          |
| checkerboard<br>texture        | transposed-conv<br>overlap       | compare<br>resize+conv at<br>same output size |
| shifted instance<br>mask       | RoI quantization                 | compare RoIPool<br>with RoIAlign              |

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| shifted instance<br>mask       | RoI quantization                 | compare RoIPool<br>with RoIAlign              |

**Inspect spatial alignment before changing the loss.**

# Architecture checkpoint: context and coordinates play different roles QUESTION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

If the deep path is removed but every high-resolution skip remains, what is most at risk?

**A**

semantic meaning

**B**

number of pixels

**C**

softmax normalization

# Architecture checkpoint: context and coordinates play different roles QUESTION

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If the deep path is removed but every high-resolution skip remains, what is most at risk?

**A**

semantic meaning

**B**

number of pixels

**C**

softmax normalization

**A. A crisp local edge does not by itself say whether the region is road, cat, or shadow.**

**INFER makes a mask; EVAL  
measures its agreement**

---

# TRAIN ends with probabilities; INFER begins without labels

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$G, p \rightarrow \mathcal{L}_{\text{BCE}}, \mathcal{L}_{\text{Dice}}$$

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**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$p, \tau \rightarrow P; \text{no truth and no loss}$$

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**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$G, p \rightarrow \mathcal{L}_{\text{BCE}}, \mathcal{L}_{\text{Dice}}$$

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$p, \tau \rightarrow P; \text{no truth and no loss}$$

**The same  $p$  crosses a clock boundary; its meaning does not change.**

# Commit: threshold the held probabilities at .50

QUESTION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

**HELD CASE · DECISION RULE**  $P_i = 1[p_i \geq .50]$ . Equality is foreground by convention.

truth  $G$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |



foreground  $p$

|     |     |     |     |
|-----|-----|-----|-----|
| .10 | .90 | .90 | .10 |
| .60 | .90 | .90 | .10 |
| .10 | .90 | .30 | .60 |
| .10 | .60 | .05 | .05 |

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**INFER** probability  $\rightarrow$  label / object masks

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foreground  $p$

|     |     |     |     |
|-----|-----|-----|-----|
| .10 | .90 | .90 | .10 |
| .60 | .90 | .90 | .10 |
| .10 | .90 | .30 | .60 |
| .10 | .60 | .05 | .05 |

Before revealing  $P$ , predict the four counts: TP, FP, FN, TN.

# Thresholding yields this exact hard mask

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

truth  $G$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |



foreground  $p$

|     |     |     |     |
|-----|-----|-----|-----|
| .10 | .90 | .90 | .10 |
| .60 | .90 | .90 | .10 |
| .10 | .90 | .30 | .60 |
| .10 | .60 | .05 | .05 |



hard  $P$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 |

# Thresholding yields this exact hard mask

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

truth  $G$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |



foreground  $p$

|     |     |     |     |
|-----|-----|-----|-----|
| .10 | .90 | .90 | .10 |
| .60 | .90 | .90 | .10 |
| .10 | .90 | .30 | .60 |
| .10 | .60 | .05 | .05 |



hard  $P$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 |

Thresholding is nondifferentiable; it belongs to inference or metric computation, not the training path.

# Count errors before naming a metric

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice**INFER** probability  $\rightarrow$  label / object masks**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP**truth**  $G$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

**prediction**  $P$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 |

# Count errors before naming a metric

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

truth  $G$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

prediction  $P$

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 |

$$\text{TP} = 5$$

# Count errors before naming a metric

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice**INFER** probability  $\rightarrow$  label / object masks**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP**truth**  $G$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

**prediction**  $P$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 |

$$\text{TP} = 5 \quad \text{FP} = 3$$

# Count errors before naming a metric

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice**INFER** probability  $\rightarrow$  label / object masks**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP**truth**  $G$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

**prediction**  $P$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 |

$$\text{TP} = 5 \quad \text{FP} = 3 \quad \text{FN} = 1$$

# Count errors before naming a metric

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice**INFER** probability  $\rightarrow$  label / object masks**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP**truth**  $G$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

**prediction**  $P$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 |

$$TP = 5 \quad FP = 3 \quad FN = 1 \quad TN = 7$$

# Count errors before naming a metric

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice**INFER** probability  $\rightarrow$  label / object masks**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP**truth**  $G$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

**prediction**  $P$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 1 | 1 | 1 | 0 |
| 0 | 1 | 0 | 1 |
| 0 | 1 | 0 | 0 |

$$TP = 5 \quad FP = 3 \quad FN = 1 \quad TN = 7$$

**Every hard metric below is a different reduction of the same four counts.**

# Pixel accuracy can reward an empty foreground prediction

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

**HELD CASE · HELD MASK** accuracy =  $\frac{TP+TN}{16} = \frac{5+7}{16} = .750$ .

# Pixel accuracy can reward an empty foreground prediction

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**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

**HELD CASE · HELD MASK**  $\text{accuracy} = \frac{TP+TN}{16} = \frac{5+7}{16} = .750.$

Now imagine a  $100 \times 100$  image with only 100 tumor pixels.

# Pixel accuracy can reward an empty foreground prediction

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Now imagine a  $100 \times 100$  image with only 100 tumor pixels.

predict background everywhere  $\rightarrow$  accuracy =  $\frac{9900}{10000} = 99\%$ .

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Now imagine a  $100 \times 100$  image with only 100 tumor pixels.

predict background everywhere  $\rightarrow \text{accuracy} = \frac{9900}{10000} = 99\%.$

tumor recall  $= \frac{0}{100} = 0.$

# Pixel accuracy can reward an empty foreground prediction

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Now imagine a  $100 \times 100$  image with only 100 tumor pixels.

predict background everywhere  $\rightarrow$  accuracy =  $\frac{9900}{10000} = 99\%$ .

tumor recall =  $\frac{0}{100} = 0$ .

**Accuracy weights the easy background by its frequency; it may not reflect the foreground task.**

# IoU removes true background from the denominator

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$\text{IoU} = \frac{|P \cap G|}{|P \cup G|} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}.$$

# IoU removes true background from the denominator

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

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**HELD CASE · SAME COUNTS**  $|P \cap G| = 5$  and  $|P \cup G| = 5 + 3 + 1 = 9$ .

# IoU removes true background from the denominator

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**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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**HELD CASE · SAME COUNTS**  $|P \cap G| = 5$  and  $|P \cup G| = 5 + 3 + 1 = 9$ .

$$\text{IoU} = \frac{5}{9} \approx .556.$$

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$$\text{IoU} = \frac{5}{9} \approx .556.$$

**Seven true-background pixels cannot inflate foreground IoU.**

# Dice gives the intersection twice the weight

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}.$$

# Dice gives the intersection twice the weight

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}.$$

**HELD CASE · SAME COUNTS**  $|P| = 8$ ,  $|G| = 6$ , and  $|P \cap G| = 5$ .

# Dice gives the intersection twice the weight

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**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

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$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}.$$

**HELD CASE · SAME COUNTS**  $|P| = 8$ ,  $|G| = 6$ , and  $|P \cap G| = 5$ .

$$\text{Dice} = \frac{10}{14} \approx .714.$$

# Dice gives the intersection twice the weight

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**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$\text{Dice} = \frac{2|P \cap G|}{|P| + |G|} = \frac{2\text{TP}}{2\text{TP} + \text{FP} + \text{FN}}.$$

**HELD CASE · SAME COUNTS**  $|P| = 8$ ,  $|G| = 6$ , and  $|P \cap G| = 5$ .

$$\text{Dice} = \frac{10}{14} \approx .714.$$

**For one binary mask, Dice is the foreground F1 score.**

# Dice is a monotone transform of IoU for one mask pair

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$\text{Dice} = \frac{2\text{IoU}}{1+\text{IoU}}, \quad \text{IoU} = \frac{\text{Dice}}{2-\text{Dice}}.$$

# Dice is a monotone transform of IoU for one mask pair

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**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$\text{Dice} = \frac{2\text{IoU}}{1+\text{IoU}}, \quad \text{IoU} = \frac{\text{Dice}}{2-\text{Dice}}.$$

$$\frac{2(\frac{5}{9})}{1 + \frac{5}{9}} = \frac{10}{14} \approx .714.$$

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$$\text{Dice} = \frac{2\text{IoU}}{1+\text{IoU}}, \quad \text{IoU} = \frac{\text{Dice}}{2-\text{Dice}}.$$

$$\frac{2(\frac{5}{9})}{1 + \frac{5}{9}} = \frac{10}{14} \approx .714.$$

Qualifier: this preserves ranking for a single class and mask pair. Macro-averaging nonlinear scores over classes or images can change a model ranking.

# Soft Dice scores the held probabilities before thresholding

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$\text{SoftDice} = \frac{2 \sum_i p_i g_i + \varepsilon}{\sum_i p_i + \sum_i g_i + \varepsilon}.$$

# Soft Dice scores the held probabilities before thresholding

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$\text{SoftDice} = \frac{2 \sum_i p_i g_i + \varepsilon}{\sum_i p_i + \sum_i g_i + \varepsilon}.$$

**HELD CASE · SAME  $G, p$**   $\sum p = 7.2, \sum g = 6, \sum pg = 4.8$ ; omit tiny  $\varepsilon$  in the displayed arithmetic.

# Soft Dice scores the held probabilities before thresholding

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**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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**HELD CASE · SAME  $G, p$**   $\sum p = 7.2, \sum g = 6, \sum pg = 4.8$ ; omit tiny  $\varepsilon$  in the displayed arithmetic.

$$\text{SoftDice} = \frac{9.6}{13.2} \approx .727.$$

# Soft Dice scores the held probabilities before thresholding

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$\text{SoftDice} = \frac{2 \sum_i p_i g_i + \varepsilon}{\sum_i p_i + \sum_i g_i + \varepsilon}.$$

**HELD CASE · SAME  $G, p$**   $\sum p = 7.2, \sum g = 6, \sum pg = 4.8$ ; omit tiny  $\varepsilon$  in the displayed arithmetic.

$$\text{SoftDice} = \frac{9.6}{13.2} \approx .727.$$

$$\mathcal{L}_{\text{Dice}} = 1 - .727 = .273.$$

# CE and Soft Dice expose different failure surfaces

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

## LOCAL CALIBRATION · CE

scores every valid pixel; strong  
gradients identify which probabilities  
are wrong

## REGION OVERLAP · DICE

couples foreground pixels through one  
region denominator; resists  
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$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{Dice}}.$$

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$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{Dice}}.$$

**Combining them does not make either definition optional: declare  $\lambda$ , reduction, and empty-mask policy.**

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice**INFER** probability  $\rightarrow$  label / object masks**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

For each class  $k$ ,

$$\text{IoU}_k = \frac{\text{TP}_k}{\text{TP}_k + \text{FP}_k + \text{FN}_k}, \quad \text{mIoU} = \frac{1}{|K_{\text{report}}|} \sum_{k \in K_{\text{report}}} \text{IoU}_k.$$

# Semantic mIoU averages declared per-class IoUs

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**CLASSES**

**VOID**

**REDUCTION**

include or exclude  
background?

remove ignored pixels  
from all counts

image mean or dataset-  
level confusion?

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**CLASSES**

**VOID**

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include or exclude  
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image mean or dataset-  
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**“mIoU” is incomplete until its class set and aggregation rule are named.**

# Diagnose a surprising score on the correct clock

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice**INFER** probability  $\rightarrow$  label / object masks**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

| Symptom                      | Suspect                        | Smallest test                        |
|------------------------------|--------------------------------|--------------------------------------|
| 99% accuracy;<br>IoU = 0     | rare foreground                | foreground recall;<br>class counts   |
| loss falls; mIoU<br>flat     | threshold or class<br>collapse | histogram $p$ by class;<br>confusion |
| good box AP;<br>poor mask AP | boundary<br>alignment          | compare box IoU<br>with mask IoU     |

# Diagnose a surprising score on the correct clock

**TRAIN** probability ↔ truth; CE / Dice

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**Loss, inference policy, and evaluation protocol are three different levers.**

**One semantic region is not yet  
two objects**

---

# The held union cannot count its touching instances

same orchard tile



semantic union

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 1 | 1 | 0 |
| 0 | 0 | 0 | 0 |

one connected component

required identities

|   |   |   |   |
|---|---|---|---|
| 0 | 1 | 2 | 0 |
| 0 | 1 | 2 | 0 |
| 0 | 1 | 2 | 0 |
| 0 | 0 | 0 | 0 |

id 1 vs id 2

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semantic union

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id 1 vs id 2

**Semantic class evidence answers area; instance identity answers count.**

# Instance segmentation returns a scored set—not one class grid

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$x \rightarrow \left\{ \left( \hat{y}_j, \hat{b}_j, \hat{m}_j, \hat{s}_j \right) \right\}_{j=1}^N.$$

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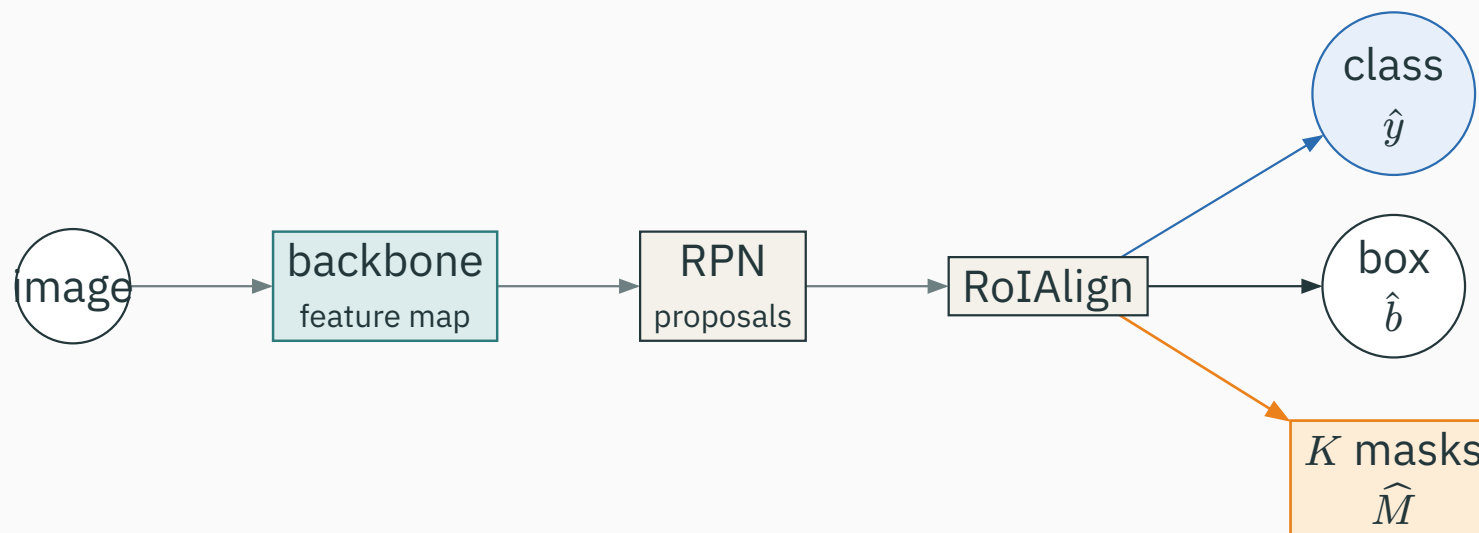
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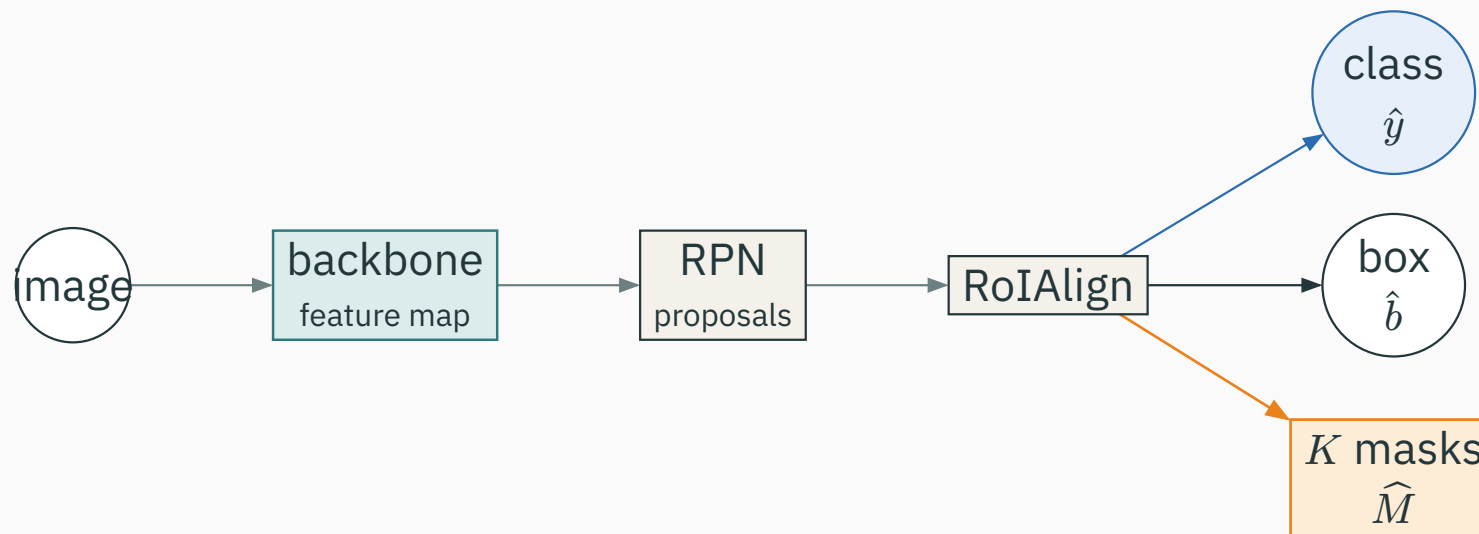
$$x \rightarrow \left\{ \left( \hat{y}_j, \hat{b}_j, \hat{m}_j, \hat{s}_j \right) \right\}_{j=1}^N.$$

| Symbol      | Shape                        | Meaning                  |
|-------------|------------------------------|--------------------------|
| $\hat{y}_j$ | scalar                       | object class             |
| $\hat{b}_j$ | four<br>coordinates          | box that places the mask |
| $\hat{m}_j$ | $m \times m$ then<br>resized | shape in the RoI frame   |
| $\hat{s}_j$ | scalar                       | ranking confidence       |

# Mask R-CNN adds a mask branch parallel to class and box

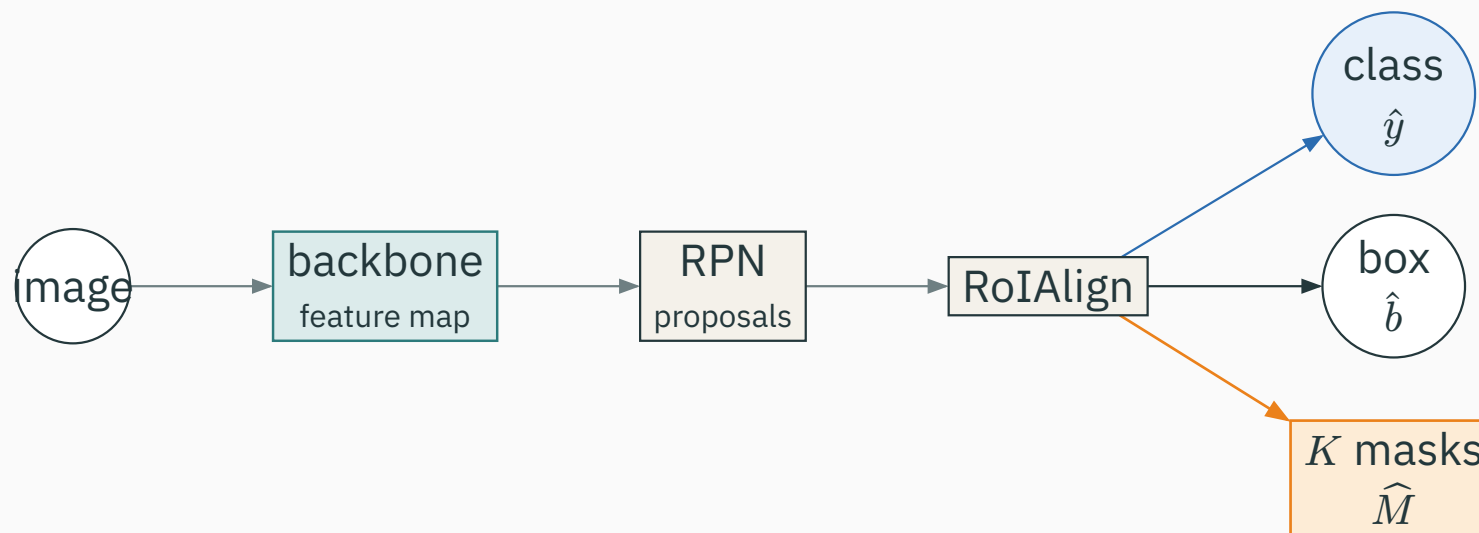


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RPN = Region Proposal Network; RoI = Region of Interest.

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RPN = Region Proposal Network; RoI = Region of Interest.

Detection proposes **which object**; the mask branch predicts **which RoI pixels**.

# Separate the RPN loss from the second-stage RoI loss

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

## STAGE 1 · RPN

$\mathcal{L}_{\text{RPN-obj}}$  on sampled anchors  
 $\mathcal{L}_{\text{RPN-box}}$  on positive anchors

## STAGE 2 · SAMPLED ROIS

$\mathcal{L}_{\text{cls}}$  on positive + negative RoIs  
 $\mathcal{L}_{\text{box}} + \mathcal{L}_{\text{mask}}$  on positive RoIs

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The familiar three-term Mask R-CNN equation is the **second-stage RoI loss**, not the whole RPN-plus-RoI total.

# Original Mask R-CNN predicts $K$ independent binary masks

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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$$\hat{M} \in \mathbb{R}^{K \times m \times m}, \quad \hat{M}_{k,u,v} = \sigma(a_{k,u,v}).$$

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softmax chooses the object category

**MASK CHANNEL  $k$**

sigmoid makes foreground/  
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A class-agnostic one-mask head is a valid later variant; it is not the original  $K$ -mask formulation.

# Calculate one selected-channel mask BCE

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

$$M = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}, \quad \hat{M} = \begin{pmatrix} .8 & .6 \\ .3 & .1 \end{pmatrix}.$$

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$$= \frac{1}{4} [.223 + .511 + .357 + .105] \approx .299.$$

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**Pixels in this RoI and the selected ground-truth class channel define the mask loss.**

## ROIPOOL

---

round RoI and bin coordinates;  
convenient, but spatially quantized



## ROIALIGN

---

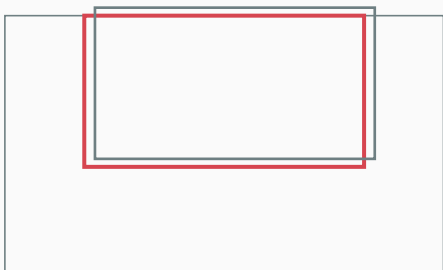
keep fractional coordinates; sample  
neighboring features bilinearly



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The Mask R-CNN paper reports box-AP gains too; the alignment benefit is larger for pixel-accurate masks.

# Instance evaluation reuses L10 AP with mask IoU

**D** DERIVATION

**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

**INFER** probability  $\rightarrow$  label / object masks

**EVAL** hard output  $\leftrightarrow$  truth; IoU / AP

score-sort  
predicted masks

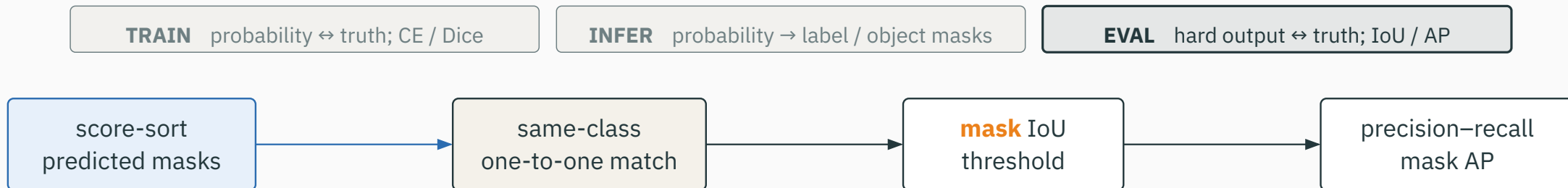
same-class  
one-to-one match

**mask** IoU  
threshold

precision-recall  
mask AP

# Instance evaluation reuses L10 AP with mask IoU

**D** DERIVATION



**Box AP asks whether rectangles overlap; mask AP asks whether object pixels overlap.**

**SHOULD: policies, alignment,  
and panoptic output**

---

# SHOULD · void pixels leave both the loss sum and denominator

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Let  $V$  be the valid labeled pixels:

$$\mathcal{L}_{\text{CE}} = -\frac{1}{|V|} \sum_{i \in V} \log p_{i, y_i}.$$

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**VALID**

include

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include

**VOID**

exclude

**VALID**

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**Remove ignored pixels from CE, confusion counts, IoU, and Dice—not only from the numerator.**

# SHOULD · empty masks and class averaging need a written policy

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**TRAIN** probability  $\leftrightarrow$  truth; CE / Dice

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raw Dice is  $\frac{0}{0}$ ; choose smoothing, exclusion, or a declared perfect score

## MULTI-CLASS

compute per class; declare background inclusion, empty-class handling, and macro or frequency weighting

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Also declare whether scores are averaged per image or accumulated from a dataset-level confusion matrix.

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**The reduction rule is part of the metric—not an implementation footnote.**

| Method                    | What creates locations?                   | What learns?                     |
|---------------------------|-------------------------------------------|----------------------------------|
| nearest /<br>bilinear     | fixed interpolation                       | following<br>convolution refines |
| transposed<br>convolution | learned overlap<br>pattern                | kernel and mixing<br>jointly     |
| skip + resize             | decoder plus<br>aligned encoder<br>detail | fusion after<br>concatenation    |

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**Every route needs evidence to refine the larger grid; interpolation alone cannot reverse pooling.**

# SHOULD · bilinear interpolation is a weighted four-neighbor sum

D DERIVATION

Suppose a RoIAlign sample lies halfway in both directions between

$$\begin{pmatrix} 1 & 3 \\ 5 & 7 \end{pmatrix}.$$

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All four weights equal .25:

$$f = .25(1) + .25(3) + .25(5) + .25(7) = 4.$$

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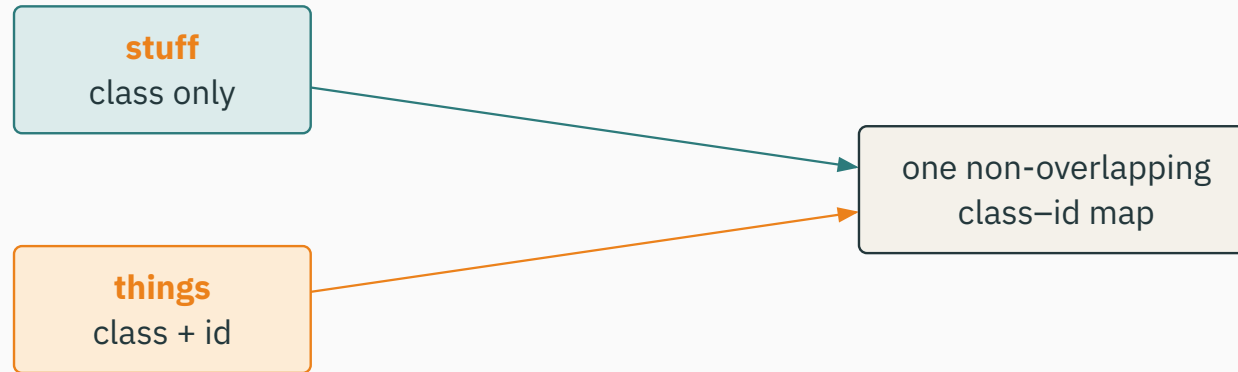
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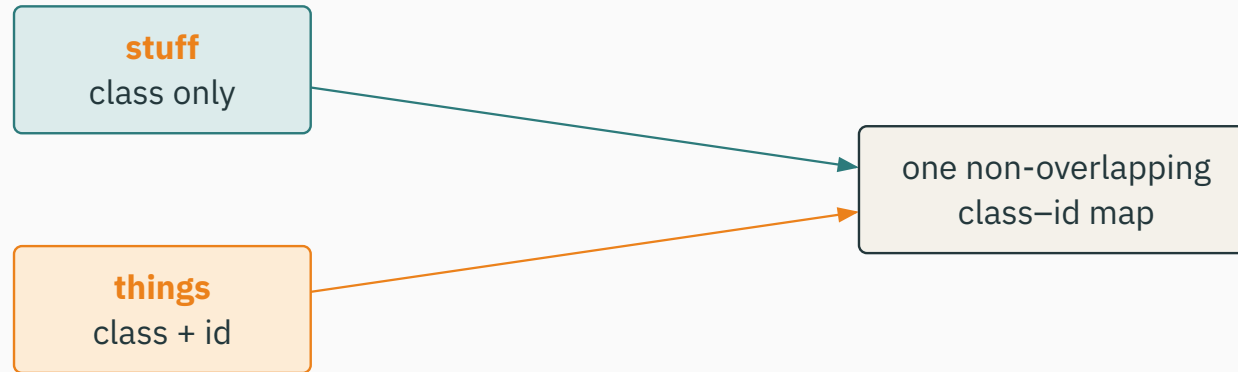
**Precise sampling preserves alignment; it does not invent image detail.**

# SHOULD · panoptic output assigns one class–identity record per pixel

V VISUAL



# SHOULD · panoptic output assigns one class–identity record per pixel



Overlapping instance hypotheses must be fused; datasets may reserve a void label for uncertain pixels.

# SHOULD · choose the output and metric from the question

| Need                 | Output                        | Evaluation                       |
|----------------------|-------------------------------|----------------------------------|
| road area            | $H \times W$ semantic classes | per-class IoU / mIoU             |
| count touching trees | scored set of instance masks  | mask AP under mask IoU           |
| whole labeled scene  | $H \times W$ class-id map     | PQ or declared panoptic protocol |

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**Architecture names follow the output contract; they do not define the task.**

# OPTIONAL · primary papers establish the architectural claims

## FCN · 2014

---

Long, Shelhamer & Darrell · pixels-to-pixels; coarse/deep plus fine/shallow fusion

## V-NET · 2016

---

Milletari, Navab & Ahmadi · Dice-based training objective for imbalance

## PANOPTIC · 2018

---

Kirillov et al. · unified stuff-and-things task and PQ

## U-NET · 2015

---

Ronneberger, Fischer & Brox · contracting context plus expanding localization

## MASK R-CNN · 2017

---

He et al. · parallel mask branch, RoIAlign, independent binary masks

## EXACT COMPANION

---

L10 toy-mask Colab · retained logits, CE, held mask, IoU, Dice, and mask BCE

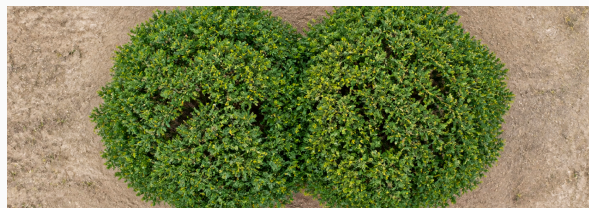
| Variant                   | What changes                                            | What stays explicit                     |
|---------------------------|---------------------------------------------------------|-----------------------------------------|
| class-agnostic mask head  | $1 \times m \times m$ rather than $K \times m \times m$ | object class comes from class head      |
| dilated / pyramid decoder | context without identical downsampling                  | output grid and pixel loss              |
| query-based mask sets     | learned queries replace RPN RoIs                        | one-to-one identity and mask metric     |
| panoptic fusion           | resolve overlaps and stuff regions                      | one class-identity assignment per pixel |

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Do not turn variants into a catalogue: ask which output, alignment, loss, or matching assumption changed.

# Revisit the orchard commitment before revealing the answer

QUESTION



**HELD CASE · SAME REQUIREMENT** Report total canopy area **and** count two touching trees.

**A**

global class

**B**

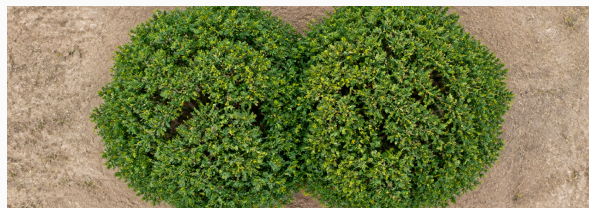
class + box

**C**

semantic map

**D**

separate object  
masks



**HELD CASE · SAME REQUIREMENT** Report total canopy area **and** count two touching trees.

**A**

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class + box

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semantic map

**D**

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**D. Instance masks preserve the two identities; their union gives total area. If every background/stuff pixel also needs a class, use a panoptic class-identity map.**

## SEGMENTATION CHECKLIST

1. What is the output structure?
2. Which axis holds classes?
3. What enters TRAIN loss?
4. Which INFER threshold/fusion applies?
5. Which EVAL classes, masks, and reductions define the score?

## PUBLIC L12 · NEXT TOKEN

Pixel prediction shared one classifier over spatial positions  $(h, w)$ .  
Next-token prediction shares one classifier over sequence positions  $t$ .  
Pixel classes become vocabulary tokens; **causality** becomes the new constraint.

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3. What enters TRAIN loss?
4. Which INFER threshold/fusion applies?
5. Which EVAL classes, masks, and reductions define the score?

## PUBLIC L12 · NEXT TOKEN

Pixel prediction shared one classifier over spatial positions  $(h, w)$ .  
Next-token prediction shares one classifier over sequence positions  $t$ .  
Pixel classes become vocabulary tokens; **causality** becomes the new constraint.

**A dense model is trustworthy only when output support, three clocks, alignment, and metric protocol agree.**